

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Reflection.Emit;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace linq
{
    public partial class WebForm2 : System.Web.UI.Page
    {
        [Serializable]
        public class student
        {
            public string name { get; set; }
            public int age { get; set; }
        }

        private List<student> stud
        {
            get
            {
                if (ViewState["stud"] == null)
                    ViewState["stud"] = new List<student>();
                return (List<student>)ViewState["stud"];
            }
        }
    }
}

```

```

    }
}

protected void Page_Load(object sender, EventArgs e)
{

}

protected void btnadd_Click(object sender, EventArgs e)
{
    string name1 = txtname.Text;
    int age1 =int.Parse(txtage.Text);

    stud.Add(new student { name = name1,age = age1 });

    lblshow.Text = "Student added successfully";
    txtname.Text = "";
    txtage.Text = "";
}

protected void btndisplay_Click(object sender, EventArgs e)
{
    lblshow.Text = "Student Details<br/>";
    foreach (var s in stud)
    {
        lblshow.Text += s.name + " " + s.age.ToString() + "<br>";
    }
}

```

```
}
```

```
protected void btnsearch_Click(object sender, EventArgs e)
```

```
{
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```
    var namesWithS = stud
```

```
    .Where(s => s.name.StartsWith("D",StringComparison.OrdinalIgnoreCase))
```

```
    .Select(s => s.name);
```

```
    lblshow.Text = "Students(starting with d):" + string.Join(", ", namesWithS);
```

```
}
```

```
protected void btnagg_Click(object sender, EventArgs e)
```

```
{
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```
    int cnt = stud.Count;
```

```
    int sum = stud.Sum(s => s.age);
```

```
    double avg = stud.Average(s => s.age);
```

```
    int max = stud.Max(s => s.age);
```

```
    int min = stud.Min(s => s.age);
```

```
    lblshow.Text = "Count : " + cnt.ToString() + "<br>Sum : " + sum.ToString() +  
"<br>Average : " + avg.ToString() + "<br>Max : " + max.ToString() + "<br>Min : " +  
min.ToString();
```

```
}
```

```
protected void btnfirst_Click(object sender, EventArgs e)
```

```
{
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```
    var fir = stud.First().name;
```

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    lblfirst.Text = "First student : "+fir.ToString();  
}
```

```
protected void btnlast_Click(object sender, EventArgs e)  
{  
    var las = stud.Last().name;  
    lbllast.Text = "Last student : " + las.ToString();  
}
```

```
protected void btndefault_Click(object sender, EventArgs e)  
{  
    foreach (var s in stud.Where(x => x.age>19))  
    {  
        lblshow.Text += s.name + " " + s.age.ToString() + "<br>";  
    }  
    var result = stud.FirstOrDefault(s => s.age>19);  
    if(result != null)  
        lbldefault.Text = result.age.ToString();  
  
    var result1 = stud.LastOrDefault(s => s.age > 19);  
    if (result1 != null)  
        lblldefault.Text = result1.age.ToString();  
}
```

```
protected void btnsingle_Click(object sender, EventArgs e)  
{
```

```
if(stud.Count==1)
    lblshow.Text=stud.Single().name;
else
    lblshow.Text = "More than 1 student";
}
}
}
```